

Large Language Models

Knowledge-base document for the Domain-Specific RAG Chatbot

1. What is a Large Language Model?

A Large Language Model (LLM) is a neural-network language model trained on large collections of text to learn statistical and semantic patterns in language. LLMs can generate text, answer questions, summarize information, transform text, and perform other language tasks.

Modern LLMs are commonly based on transformer architectures. They process tokenized input and use learned parameters to predict or generate subsequent tokens.

2. Tokens

A token is a unit used by a language model to represent input and output text. A token may correspond to a whole word, part of a word, punctuation, or another text unit depending on the tokenizer.

Before text is processed by an LLM, it is converted into a sequence of tokens. The model operates on these token representations rather than directly on raw characters.

3. Parameters

Parameters are learned numerical values inside a neural network. During training, model parameters are adjusted so that the model becomes better at its training objective.

The number of parameters is one indicator of model size, but parameter count alone does not determine the quality of every response.

4. Training Stages

Pre-training exposes the model to large amounts of text and teaches broad language patterns. A common objective for autoregressive language models is predicting the next token from preceding context.

Fine-tuning is an additional training stage in which a model is adapted to a particular dataset, task, behavior, or instruction-following objective.

Inference is the stage in which a trained model is used to generate an output for a new input.

5. Context Window

The context window is the amount of tokenized information that a model can consider in a single interaction. The available context can include the user question, conversation history, system instructions, and retrieved document passages.

A retrieval system is useful because it can select only the most relevant document passages instead of placing an entire large document collection into the model context.

6. Common LLM Capabilities

- Question answering and conversational dialogue.

- Summarization and information extraction.
- Text classification and transformation.
- Code generation and explanation.
- Drafting and rewriting natural-language content.

7. LLM Limitations

LLMs can generate fluent text that is not necessarily factually correct. They may also fail when information is missing, ambiguous, or outside the available context.

Grounding an LLM with retrieved evidence is one approach to reducing unsupported responses. In a RAG application, the retrieved passages become the evidence used for answer generation.